

Forms of Maxwell's Equation (General Expression or Dynamic or Time-Varying)

Integral Form

Point form

$$\oint_S \vec{E} \circ \hat{n} da = q_{\text{in}}/\epsilon_0 \xrightarrow[\text{theorem}]{\text{Divergence}} \vec{\nabla} \circ \vec{E} = \rho/\epsilon_0 \quad \text{Gauss's law for electric fields}$$

$$\oint_S \vec{B} \circ \hat{n} da = 0 \xrightarrow[\text{theorem}]{\text{Divergence}} \vec{\nabla} \circ \vec{B} = 0 \quad \text{Gauss's law for magnetic fields}$$

$$\oint_C \vec{E} \circ d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \circ \hat{n} da \xrightarrow[\text{theorem}]{\text{Stokes'}} \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{Faraday's law}$$

$$\oint_C \vec{B} \circ d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d}{dt} \int_S \vec{E} \circ \hat{n} da \right) \xrightarrow[\text{theorem}]{\text{Stokes'}} \vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \quad \text{Ampere-Maxwell law}$$

*in a source-free region $\rho = 0$ and $\vec{J} = 0$

** for static $\frac{d}{dt} = 0$

Recap

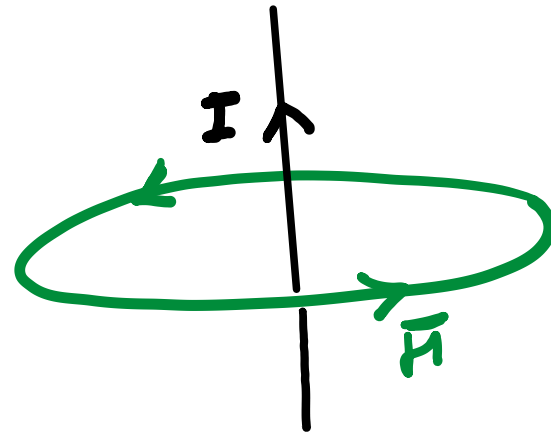
- In Maxwell's Equations, you'll encounter two kinds of electric field: the **electrostatic field** produced by electric charge and the **induced electric field** produced by a changing magnetic field.
- Similarly, there are two fundamental types of magnetic fields: **Static (or Permanent) Magnetic Field**: This type of magnetic field is associated with permanent magnets or steady currents and **Induced (or Time-Varying) Magnetic Field**: This type of magnetic field is produced by changing electric fields, according to the amended Ampère's law, which incorporates the concept of displacement current.

Ampere's Law (1820)

- For thousands of years, the only known sources of magnetic fields were certain iron cores and other materials that had been accidentally or deliberately magnetized
- In 1820, French physicist Andre-Marie Ampere heard that in Denmark Hans Christian Oersted had deflected a compass needle by passing an electric current nearby.
- Within a week, Ampere had begun quantifying the relationship between electric currents and magnetic fields (static)

AMPERE'S LAW

$$\oint \vec{H} \cdot d\vec{l} = I$$



Ampere-Maxwell's Law (1865)

- In 1865, it was Maxwell's addition of another source term (displacement current or changing electric flux) that extended Ampere's law to time-dependent conditions
- It was the presence of this term in the equation now called Ampere-Maxwell Law that allowed Maxwell to discern the electromagnetic nature of light and to develop a comprehensive theory of electromagnetism

$$\oint \vec{H} \cdot d\vec{\ell} = I + \underbrace{\int \int_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}}_{I_d}$$

NEW TERM: DISPLACEMENT CURRENT

Displacement Current

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_c + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$$

Define the displacement current as:

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{s} = \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}, \quad (6.44)$$

The displacement current does not involve real charges;
it is an equivalent current that depends on $\partial \mathbf{D} / \partial t$

where $\mathbf{J}_d = \partial \mathbf{D} / \partial t$ represents a *displacement current density*.
In view of Eq. (6.44),

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_c + I_d = I, \quad (6.45)$$

Capacitor Circuit

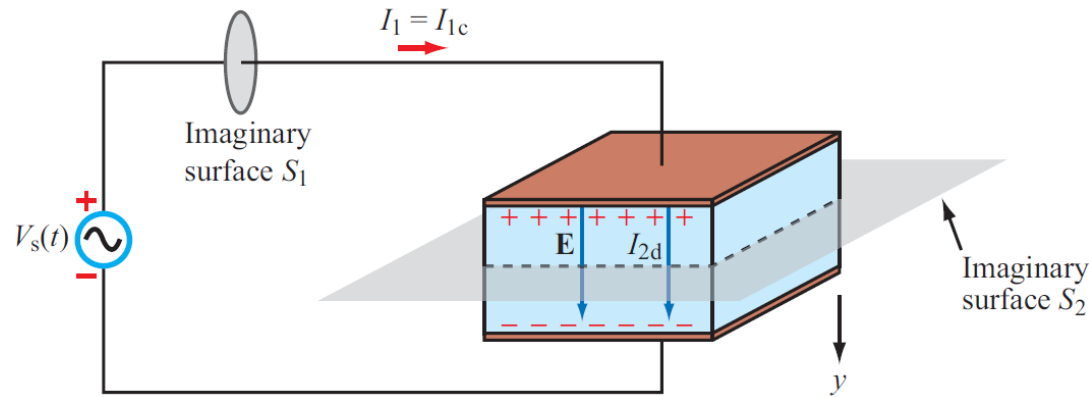
Given: Wires are perfect conductors and capacitor insulator material is perfect dielectric.

For Surface S_1 :

$$I_1 = I_{1c} + I_{1d}$$

$$I_{1c} = C \frac{dV_C}{dt} = C \frac{d}{dt} (V_0 \cos \omega t) = -C V_0 \omega \sin \omega t$$

$$I_{1d} = 0 \quad (\mathbf{D} = 0 \text{ in perfect conductor})$$



For Surface S_2 :

$$I_2 = I_{2c} + I_{2d}$$

$$I_{2c} = 0 \text{ (perfect dielectric)}$$

$$\mathbf{E} = \hat{\mathbf{y}} \frac{V_C}{d} = \hat{\mathbf{y}} \frac{V_0}{d} \cos \omega t$$

$$I_{2d} = \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$$

$$= \int_A \left[\frac{\partial}{\partial t} \left(\hat{\mathbf{y}} \frac{\epsilon V_0}{d} \cos \omega t \right) \right] \cdot (\hat{\mathbf{y}} d\mathbf{s})$$

$$= -\frac{\epsilon A}{d} V_0 \omega \sin \omega t = -C V_0 \omega \sin \omega t$$

Conclusion: $I_1 = I_2$

Displacement Current properties

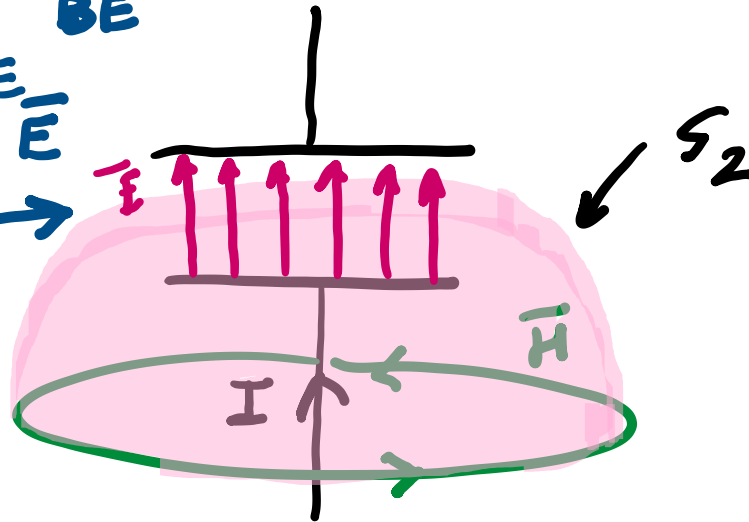
- I_d HAS THE UNITS OF CURRENT
- $I_d = 0$ FOR THE STATIC CASE.

• I_d HAS TO BE RELATED TO THE ELECTRIC FIELD $\vec{E}$ IN CAPACITOR

IN FACT

$$I_d = \iint \frac{\partial \vec{D}}{\partial t} \cdot d\vec{s}$$

$$\vec{D} = \epsilon \vec{E}$$



Example 6-7: Displacement Current Density

The conduction current flowing through a wire with conductivity $\sigma = 2 \times 10^7$ S/m and relative permittivity $\epsilon_r = 1$ is given by $I_c = 2 \sin \omega t$ (mA). If $\omega = 10^9$ rad/s, find the displacement current.

Solution: The conduction current $I_c = JA = \sigma EA$, where A is the cross section of the wire. Hence,

$$\begin{aligned} E &= \frac{I_c}{\sigma A} = \frac{2 \times 10^{-3} \sin \omega t}{2 \times 10^7 A} \\ &= \frac{1 \times 10^{-10}}{A} \sin \omega t \quad (\text{V/m}). \end{aligned}$$

$$I_d = \int_S \mathbf{J}_d \cdot d\mathbf{s} = \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}, \quad (6.44)$$

Application of Eq. (6.44), with $D = \epsilon E$, leads to

$$\begin{aligned} I_d &= J_d A \\ &= \epsilon A \frac{\partial E}{\partial t} \\ &= \epsilon A \frac{\partial}{\partial t} \left(\frac{1 \times 10^{-10}}{A} \sin \omega t \right) \\ &= \epsilon \omega \times 10^{-10} \cos \omega t = 0.885 \times 10^{-12} \cos \omega t \quad (\text{A}), \end{aligned}$$

where we used $\omega = 10^9$ rad/s and $\epsilon = \epsilon_0 = 8.85 \times 10^{-12}$ F/m. Note that I_c and I_d are in phase quadrature (90° phase shift between them). Also, I_d is about nine orders of magnitude smaller than I_c , which is why the displacement current usually is ignored in good conductors.

Charge Current Continuity Equation

Current I out of a volume is equal to rate of decrease of charge Q contained in that volume:

$$I = -\frac{dQ}{dt} = -\frac{d}{dt} \int_V \rho_v dV$$

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{d}{dt} \int_V \rho_v dV$$

$$\oint_S \mathbf{J} \cdot d\mathbf{s} = \int_V \nabla \cdot \mathbf{J} dV = -\frac{d}{dt} \int_V \rho_v dV$$

Used Divergence Theorem

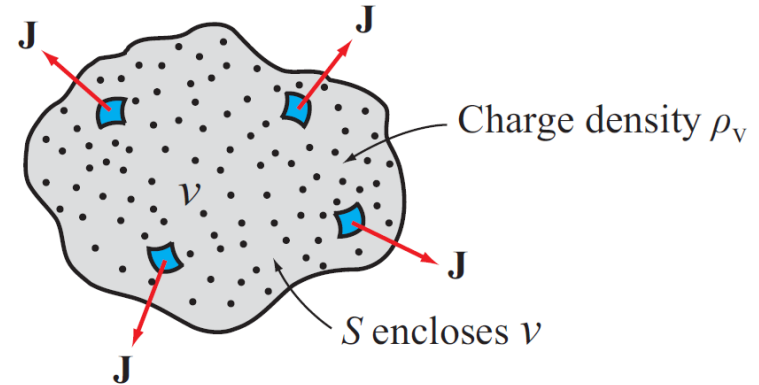


Figure 6-14: The total current flowing out of a volume V is equal to the flux of the current density $\mathbf{J}$ through the surface S , which in turn is equal to the rate of decrease of the charge enclosed in V .

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}, \quad (6.54)$$

which is known as the *charge-current continuity relation*, or simply the *charge continuity equation*.

Dielectric-Dielectric Boundary Conditions

ELECTRIC FIELDS :

- TANGENTIAL $\vec{E}$ IS CONTINUOUS

$$\vec{E}_{1t} = \vec{E}_{2t}$$

- NORMAL $\vec{D}$ IS NOT CONTINUOUS

$$\vec{D}_{1n} - \vec{D}_{2n} = \rho_s \text{ [SURFACE CHARGE DENSITY]}$$

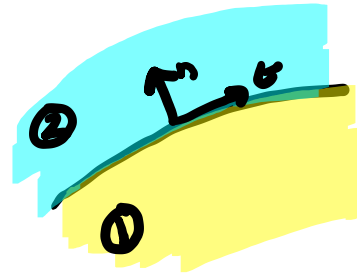
MAGNETIC FIELDS :

- TANGENTIAL $\vec{H}$ IS CONTINUOUS

$$\vec{H}_{1t} = \vec{H}_{2t}$$

- NORMAL $\vec{B}$ IS CONTINUOUS

$$\vec{B}_{1n} = \vec{B}_{2n}$$



Dielectric-Conductor Boundary Conditions

ELECTRIC FIELDS :

- NO TANGENTIAL $\vec{E}$

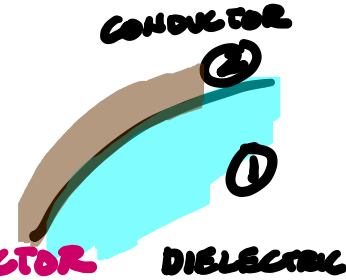
$$\vec{E}_{1t} = \vec{E}_{2t} = 0$$

- NO $\vec{E}$ IN THE CONDUCTOR

$$\vec{E}_{2n} = 0 = \vec{E}_{2t}$$

- NORMAL $\vec{D}$ IN DIELECTRIC EQUALS SURFACE CHARGE DENSITY

$$\vec{D}_{1n} = q_s$$



MAGNETIC FIELDS :

- NO MAGNETIC FIELDS IN CONDUCTOR

$$\vec{H}_{2t} = \vec{H}_{2n} = 0$$

- NO NORMAL $\vec{B}$ ANYWHERE

$$\vec{B}_{1n} = \vec{B}_{2n} = 0$$

- TANGENTIAL $\vec{H}$ IN DIELECTRIC
SURFACE CURRENT DENSITY

$$|\vec{H}_{1t}| = |J_s|$$

Summary of BC's

Table 6-2: Boundary conditions for the electric and magnetic fields.

Field Components	General Form	Medium 1 Dielectric	Medium 2 Dielectric	Medium 1 Dielectric	Medium 2 Conductor
Tangential E	$\hat{n}_2 \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$	$E_{1t} = E_{2t}$		$E_{1t} = E_{2t} = 0$	
Normal D	$\hat{n}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$	$D_{1n} - D_{2n} = \rho_s$		$D_{1n} = \rho_s$	$D_{2n} = 0$
Tangential H	$\hat{n}_2 \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$	$H_{1t} = H_{2t}$		$H_{1t} = J_s$	$H_{2t} = 0$
Normal B	$\hat{n}_2 \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0$	$B_{1n} = B_{2n}$		$B_{1n} = B_{2n} = 0$	
Notes: (1) ρ_s is the surface charge density at the boundary; (2) $\mathbf{J}_s$ is the surface current density at the boundary; (3) normal components of all fields are along $\hat{n}_2$, the outward unit vector of medium 2; (4) $E_{1t} = E_{2t}$ implies that the tangential components are equal in magnitude and parallel in direction; (5) direction of $\mathbf{J}_s$ is orthogonal to $(\mathbf{H}_1 - \mathbf{H}_2)$.					

RECALL: Phasor Domain

$$v(t) = V_0 \cos(\omega t + \phi) \\ = \Re[\underbrace{V_0 e^{j\phi}}_{\text{Phasor counterpart of } v(t)} e^{j\omega t}]$$

Phasor counterpart of $v(t)$

Time Domain

$$v(t) = V_0 \cos \omega t$$

$$v(t) = V_0 \cos(\omega t + \phi)$$

Phasor Domain

$$\longleftrightarrow \mathbf{V} = V_0$$

$$\longleftrightarrow \mathbf{V} = V_0 e^{j\phi}.$$

If $\phi = -\pi/2$,

$$v(t) = V_0 \cos(\omega t - \pi/2) \quad \longleftrightarrow \quad \mathbf{V} = V_0 e^{-j\pi/2}.$$

RECALL: Phasor Domain

$x(t)$		$\mathbf{X}$
$A \cos \omega t$	$\longleftrightarrow$	A
$A \cos(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j\phi}$
$-A \cos(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j(\phi \pm \pi)}$
$A \sin \omega t$	$\longleftrightarrow$	$Ae^{-j\pi/2} = -jA$
$A \sin(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j(\phi - \pi/2)}$
$-A \sin(\omega t + \phi)$	$\longleftrightarrow$	$Ae^{j(\phi + \pi/2)}$
$\frac{d}{dt}(x(t))$	$\longleftrightarrow$	$j\omega \mathbf{X}$
$\frac{d}{dt}[A \cos(\omega t + \phi)]$	$\longleftrightarrow$	$j\omega Ae^{j\phi}$
$\int x(t) dt$	$\longleftrightarrow$	$\frac{1}{j\omega} \mathbf{X}$
$\int A \cos(\omega t + \phi) dt$	$\longleftrightarrow$	$\frac{1}{j\omega} Ae^{j\phi}$

It is much easier to deal with exponentials in the phasor domain than sinusoidal relations in the time domain

Just need to track magnitude/phase, knowing that everything is at frequency ω

Phasor domain for Waves

$A(z, t)$	$\tilde{A}$
$A(z, t) = A \cos (\omega t - \beta z) = RE\{\textcolor{red}{A}e^{j\omega t}\textcolor{red}{e}^{-j\beta z}\}$	$\tilde{A} = Ae^{-j\beta z}$
$A \cos (\omega t + \beta z)$	$Ae^{+j\beta z}$
$A \cos (\omega t + \beta z + \varphi_o)$	$Ae^{+j\beta z}e^{+j\varphi_o}$
$Ae^{-\alpha z} \cos (\omega t + \beta z + \varphi_o)$	$Ae^{-\alpha z}e^{+j\beta z}e^{+j\varphi_o}$
$\frac{d}{dt}(A \cos (\omega t + \beta z))$	$j\omega Ae^{+j\beta z}$

Time Harmonic E & H Fields

SAY WE HAVE A TIME HARMONIC
 $\bar{E}$ FIELD : $\bar{E}(x, y, z; t)$

$$\bar{E}(x, y, z; t) = \text{Re} \left[\underbrace{\tilde{E}(x, y, z)}_{\text{PHASOR}} e^{j\omega t} \right]$$

$$\bar{H}(x, y, z; t) = \text{Re} \left[\underbrace{\tilde{H}(x, y, z)} e^{j\omega t} \right]$$

$$k = \frac{\omega}{u_p}$$

is called wavenumber or phase constant for lossless media
In general, the phase constant is denoted by β

Maxwell's Equations in Phasor form

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \cdot \tilde{E} = \rho_v / \epsilon$$

Gauss's law for electric fields

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot \tilde{B} = 0$$

Gauss's law for magnetic fields

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\begin{aligned} \vec{\nabla} \times \tilde{E} &= -j\omega \tilde{B} \\ &= -j\omega \mu \tilde{H} \\ [\tilde{B} &= \mu \tilde{H}] \end{aligned}$$

Faraday's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\begin{aligned} \vec{\nabla} \times \tilde{H} &= \tilde{J} + j\omega \tilde{D} \\ &= \tilde{J} + j\omega \epsilon \tilde{E} \end{aligned}$$



Ampere-Maxwell law

Example 6-8: Relating $\mathbf{E}$ to $\mathbf{H}$

In a nonconducting medium with $\varepsilon = 16\varepsilon_0$ and $\mu = \mu_0$, the electric field intensity of an electromagnetic wave is

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 10 \sin(10^{10}t - kz) \quad (\text{V/m}). \quad (6.88)$$

Determine the associated magnetic field intensity $\mathbf{H}$ and find the value of k .

Solution: We begin by finding the phasor $\tilde{\mathbf{E}}(z)$ of $\mathbf{E}(z, t)$. Since $\mathbf{E}(z, t)$ is given as a sine function and phasors are defined in this book with reference to the cosine function, we rewrite Eq. (6.88) as

$$\begin{aligned} \mathbf{E}(z, t) &= \hat{\mathbf{x}} 10 \cos(10^{10}t - kz - \pi/2) \quad (\text{V/m}) \\ &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right], \end{aligned} \quad (6.89)$$

with $\omega = 10^{10}$ (rad/s) and

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} 10 e^{-jkz} e^{-j\pi/2} = -\hat{\mathbf{x}} j 10 e^{-jkz}. \quad (6.90)$$

Cont.

To find both $\tilde{\mathbf{H}}(z)$ and k , we will perform a “circle”: we will use the given expression for $\tilde{\mathbf{E}}(z)$ in Faraday’s law to find $\tilde{\mathbf{H}}(z)$; then we will use $\tilde{\mathbf{H}}(z)$ in Ampère’s law to find $\tilde{\mathbf{E}}(z)$, which we will then compare with the original expression for $\tilde{\mathbf{E}}(z)$; and the comparison will yield the value of k . Application of Eq. (6.87) gives

$$\begin{aligned}
 \tilde{\mathbf{H}}(z) &= -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}} \\
 &= -\frac{1}{j\omega\mu} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -j10e^{-jkz} & 0 & 0 \end{vmatrix} \\
 &= -\frac{1}{j\omega\mu} \left[\hat{\mathbf{y}} \frac{\partial}{\partial z} (-j10e^{-jkz}) \right] \\
 &= -\hat{\mathbf{y}} j \frac{10k}{\omega\mu} e^{-jkz}. \tag{6.91}
 \end{aligned}$$

Cont.

Example 6-8 cont.

So far, we have used Eq. (6.90) for $\tilde{\mathbf{E}}(z)$ to find $\tilde{\mathbf{H}}(z)$, but k remains unknown. To find k , we use $\tilde{\mathbf{H}}(z)$ in Eq. (6.86) to find $\tilde{\mathbf{E}}(z)$:

$$\begin{aligned}\tilde{\mathbf{E}}(z) &= \frac{1}{j\omega\epsilon} \nabla \times \tilde{\mathbf{H}} \\ &= \frac{1}{j\omega\epsilon} \left[-\hat{\mathbf{x}} \frac{\partial}{\partial z} \left(-j \frac{10k}{\omega\mu} e^{-jkz} \right) \right] \\ &= -\hat{\mathbf{x}} j \frac{10k^2}{\omega^2\mu\epsilon} e^{-jkz}.\end{aligned}\tag{6.92}$$

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} 10e^{-jkz} e^{-j\pi/2} = -\hat{\mathbf{x}} j 10e^{-jkz}.\tag{6.90}$$

Equating Eqs. (6.90) and (6.92) leads to

$$k^2 = \omega^2\mu\epsilon,$$

or

$$\begin{aligned}k &= \omega\sqrt{\mu\epsilon} \\ &= 4\omega\sqrt{\mu_0\epsilon_0} \\ &= \frac{4\omega}{c} = \frac{4 \times 10^{10}}{3 \times 10^8} = 133 \quad (\text{rad/m}).\end{aligned}\tag{6.93}$$

Cont.

Example 6-8 cont.

With k known, the instantaneous magnetic field intensity is then given by

$$\begin{aligned}\mathbf{H}(z, t) &= \Re \left[\tilde{\mathbf{H}}(z) e^{j\omega t} \right] \\ &= \Re \left[-\hat{\mathbf{y}} j \frac{10k}{\omega\mu} e^{-jkz} e^{j\omega t} \right] \\ &= \hat{\mathbf{y}} 0.11 \sin(10^{10}t - 133z) \quad (\text{A/m}). \quad (6.94)\end{aligned}$$

We note that k has the same expression as the phase constant of a lossless transmission line [Eq. (2.49)].